

docking – had become routine, and space doctors had gained confidence that humans could live, work and stay healthy in space for days or even weeks at a time. Gemini also completed a long list of on-board science experiments, including studies of the space environment and Earth photography. Above all, the program added nearly 1,000 hours of valuable space-flight experience in the years between Mercury and Apollo, which was nearing flight readiness by 1966, when the Gemini programme was wrapping up. Five days before the launch of the last Gemini, Lunar Orbiter 2 had been sent to the Moon, already scouting out Apollo landing sites.

Project Apollo

Project Apollo's goals went beyond landing Americans on the Moon and returning them safely to Earth. They included 1) establishing the technology to meet other national interests in space; 2) achieving pre-eminence in space for the United States; 3) carrying out a program of scientific exploration of the moon, 4) developing man's capability to work in the lunar environment.

The docking abilities of the Gemini programme were one of the most important parts of the puzzle in figuring out how to get man on the moon. NASA had considered various different ways to get the astronauts to Earth's natural satellite, including a direct flight to the moon in a single rocket as well as a Lunar Surface Rendezvous, where two spacecraft would be launched to the moon simultaneously – one manned and one unmanned, with the latter carrying material to make the return possible.

But after much deliberation, it was finally decided that NASA would use a Lunar Orbit Rendezvous model of mission, in which two docked spacecraft were launched from Earth. They would travel to the orbit of the moon, upon which one part of the spacecraft would remain in orbit while the astronauts use the other part to make their way to the moon. This idea made it easier for the spacecraft to launch again from the moon (since it would now be a smaller and lighter machine), and once it docked with the orbiting add-on, it would get the extra power needed to return home.

The flight mode, Lunar Orbit Rendezvous, was selected in 1962. The boosters for the program were the Saturn IB for Earth orbit flights and the Saturn V for lunar flights.

Apollo was a three-part spacecraft: the Command Module (CM), which housed the crew, spacecraft operations systems, and re-entry equipment; the Service Module (SM) which carried most of the consumables (oxygen, water, helium, fuel cells, and fuel) and the main propulsion system (when